Smart Disaster Response System

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Problem Statement

Disaster situations like floods, earthquakes, and urban emergencies often face poor coordination and communication between citizens, government agencies, healthcare providers, and NGOs, which leads to delays, inefficient use of resources, and greater loss of life and property, highlighting the need for a centralized, role-based disaster management system.

Existing Issues:

- There is no centralized platform to connect all stakeholders, making coordination difficult.
- Communication is mostly manual and scattered, which causes delays.
- There is no real-time tracking of requests and resources.
- Organizations do not coordinate well during emergencies.
- There is little use of analytics and reporting.
- Emergency cases are not prioritized properly, leading to delays.

Proposed Solution:

- A centralized system is introduced to connect all stakeholders on one platform.
- Role-based access is provided so each user can perform specific tasks.
- The system enables real-time communication and better coordination.
- A work request module helps in creating, assigning, and tracking requests.
- It allows different organizations to work together smoothly.
- A reporting feature helps in analyzing performance and improving decisions.

System Architecture Overview:

Centralized Disaster Management Network

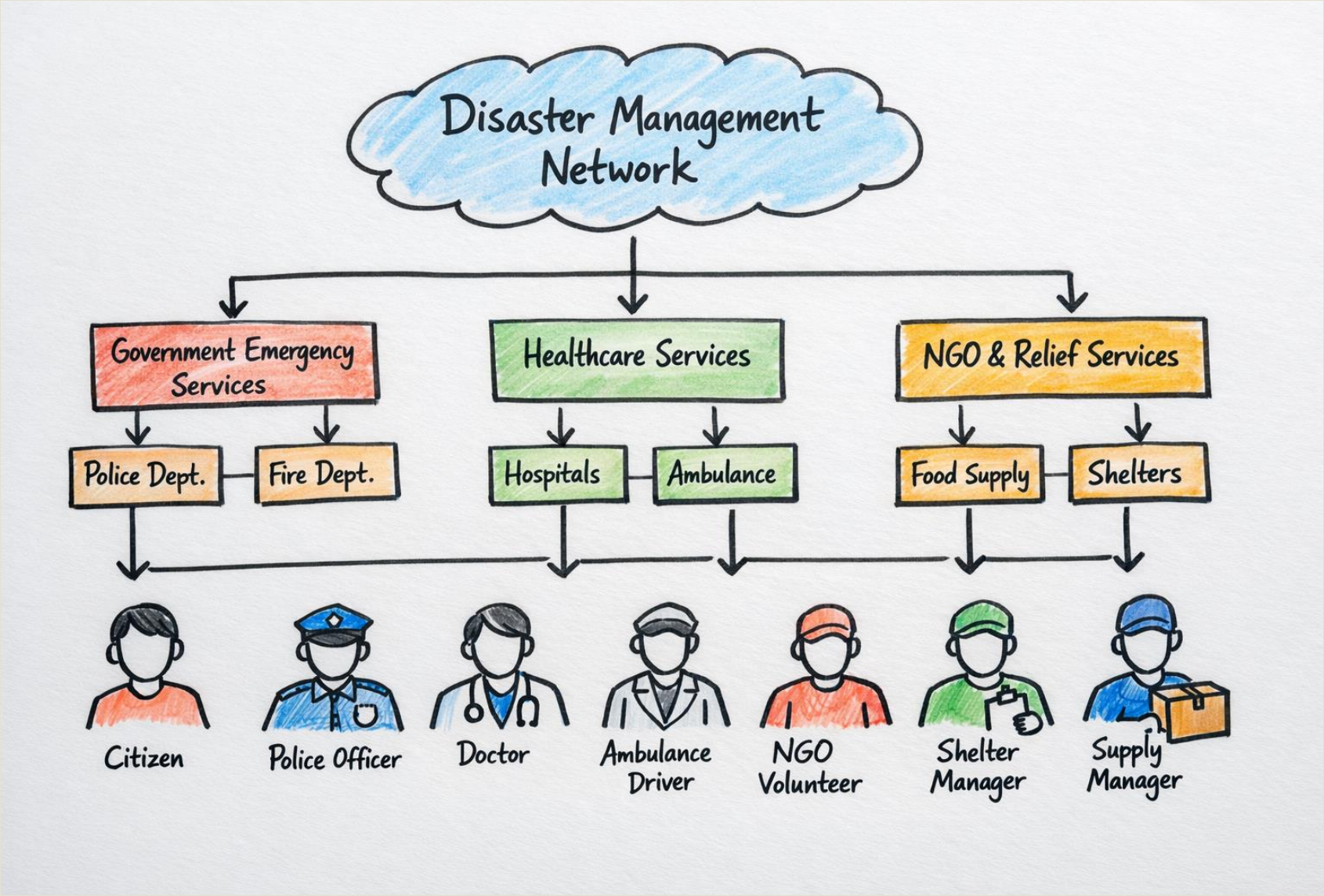
Three main enterprises:

- **Government Emergency Services**
- **Healthcare Services**
- **NGO & Relief Services**

Hierarchical structure:

- **Network → Enterprise → Organization → Users**

System Architecture Network Diagram:



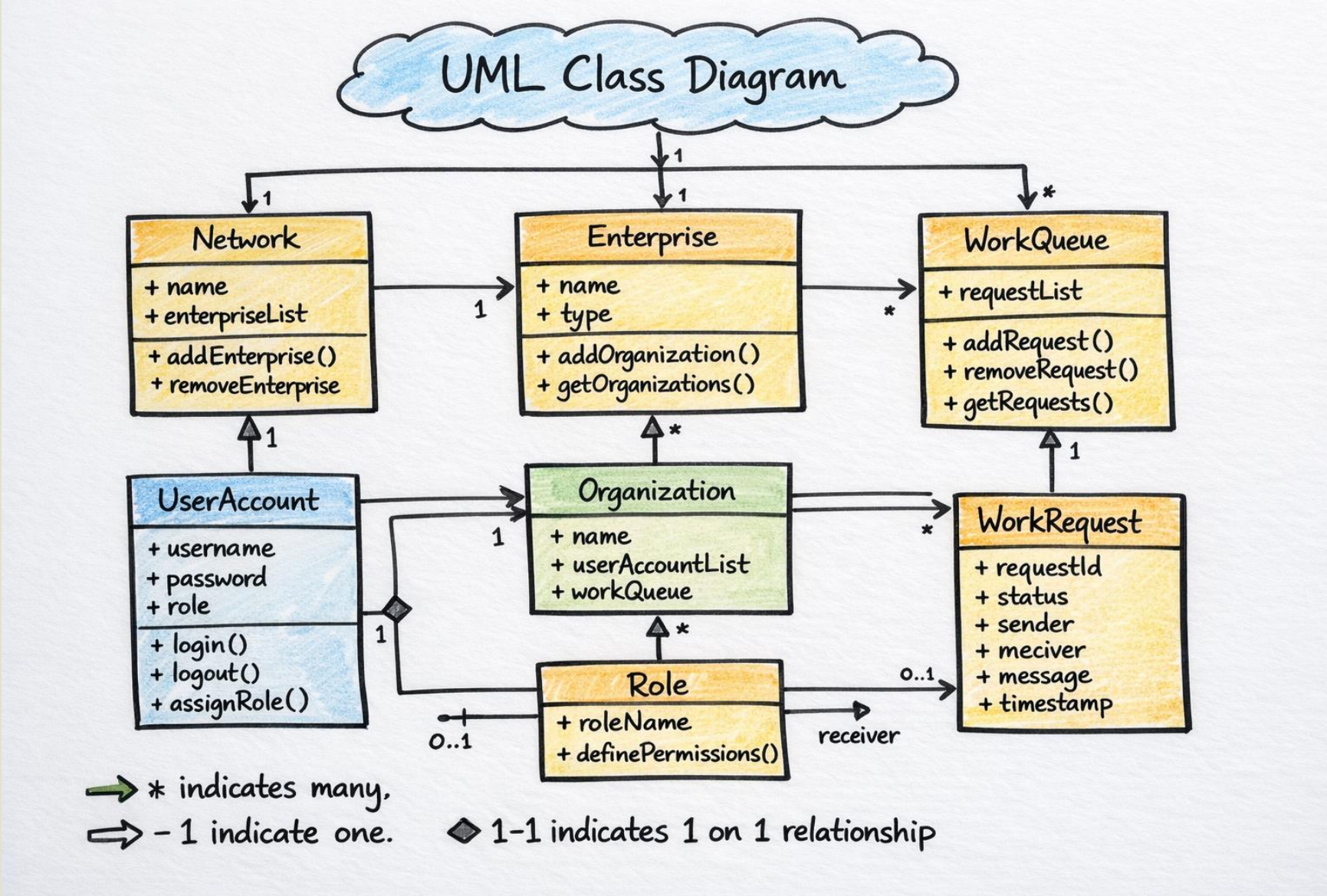
Enterprises & Organizations:

- **Government Emergency Services**
  - Police Department
  - Fire Department
- **Healthcare Services**
  - Hospitals
  - Ambulance Services
- **NGO & Relief Services**
  - Food Supply
  - Shelter Management

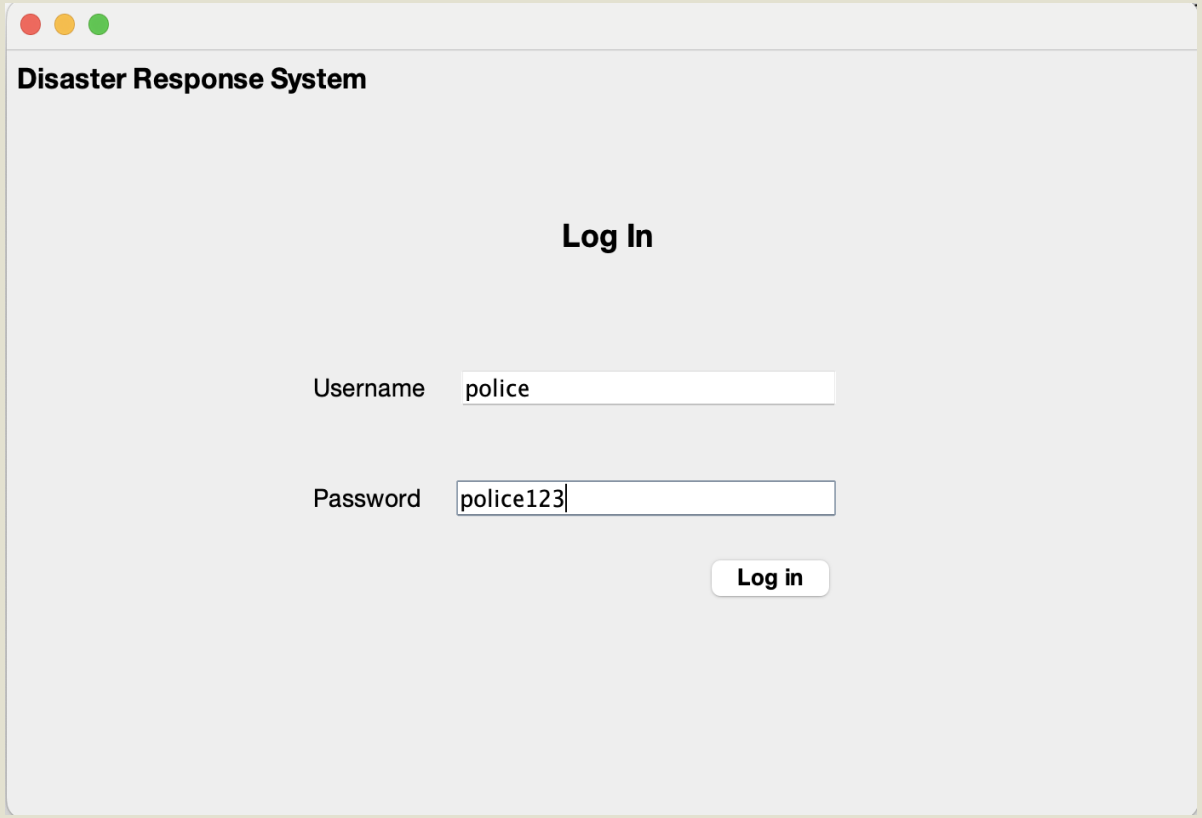
User Roles:

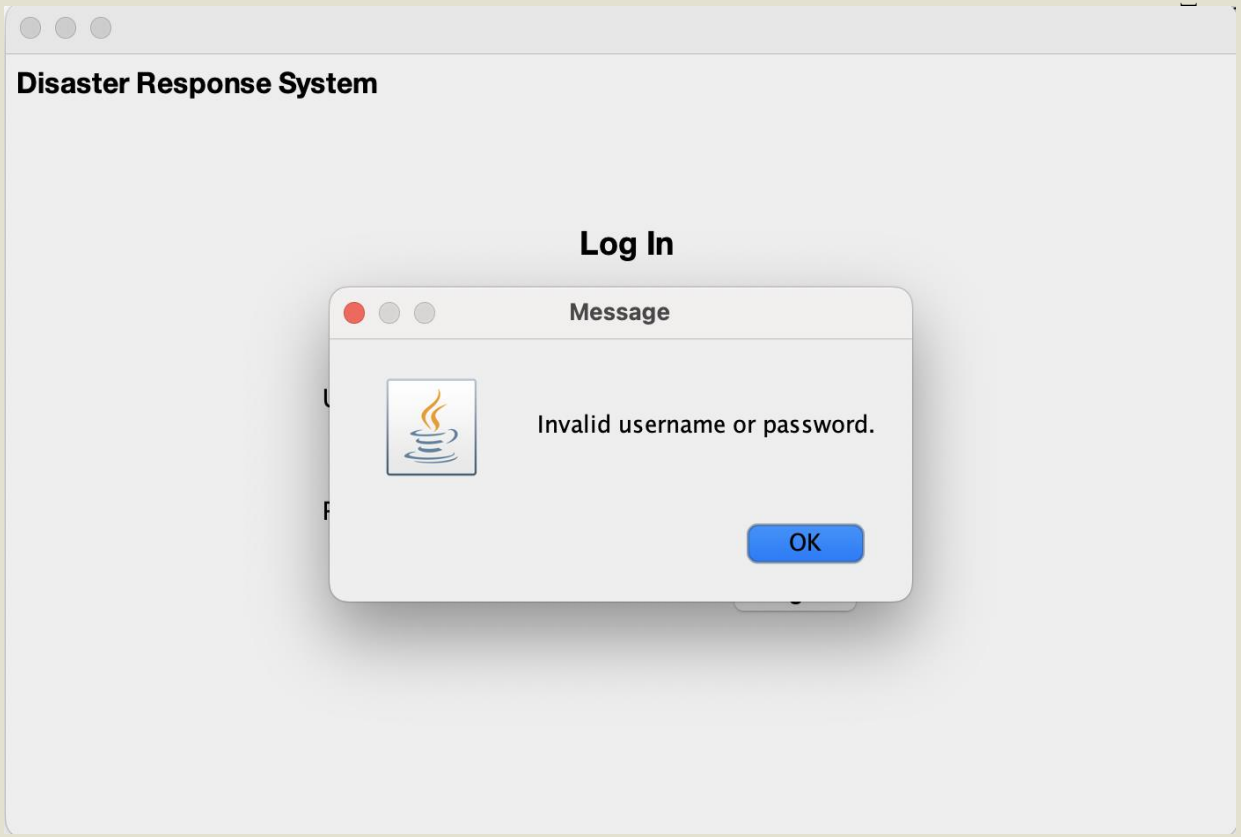
- **Citizen** – Reports emergencies and requests help during disasters.
- **Police Officer** – Maintains law and order and handles rescue coordination.
- **Fire Officer** – Manages fire-related emergencies and rescue operations.
- **Doctor** – Provides medical treatment and emergency healthcare support.
- **Ambulance Driver** – Transports injured people to hospitals quickly.
- **NGO Volunteer** – Helps in relief activities like food distribution and support.
- **Shelter Manager** – Manages temporary shelters for affected people.
- **Supply Manager** – Handles distribution of essential resources like food and supplies.

UML Relationship Diagram:

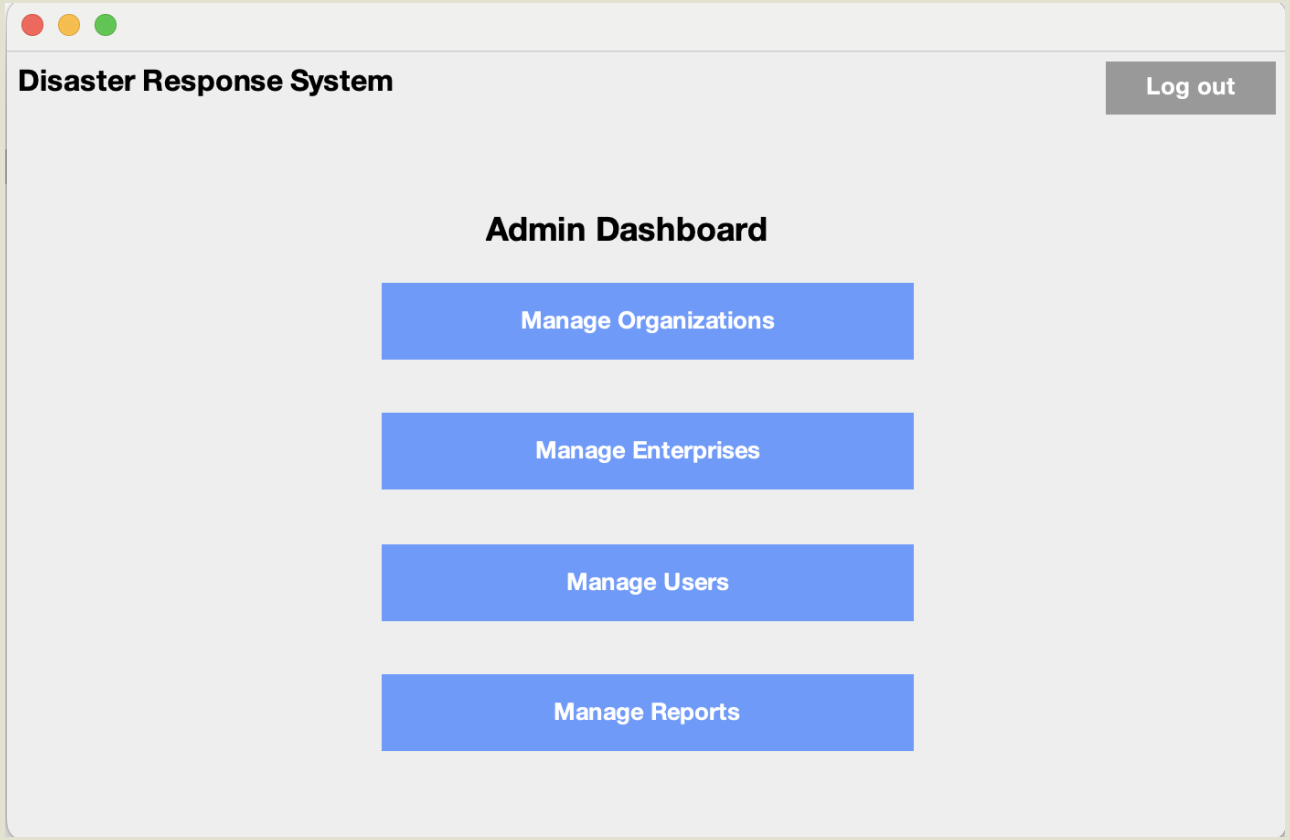


Role-Based Login Panel

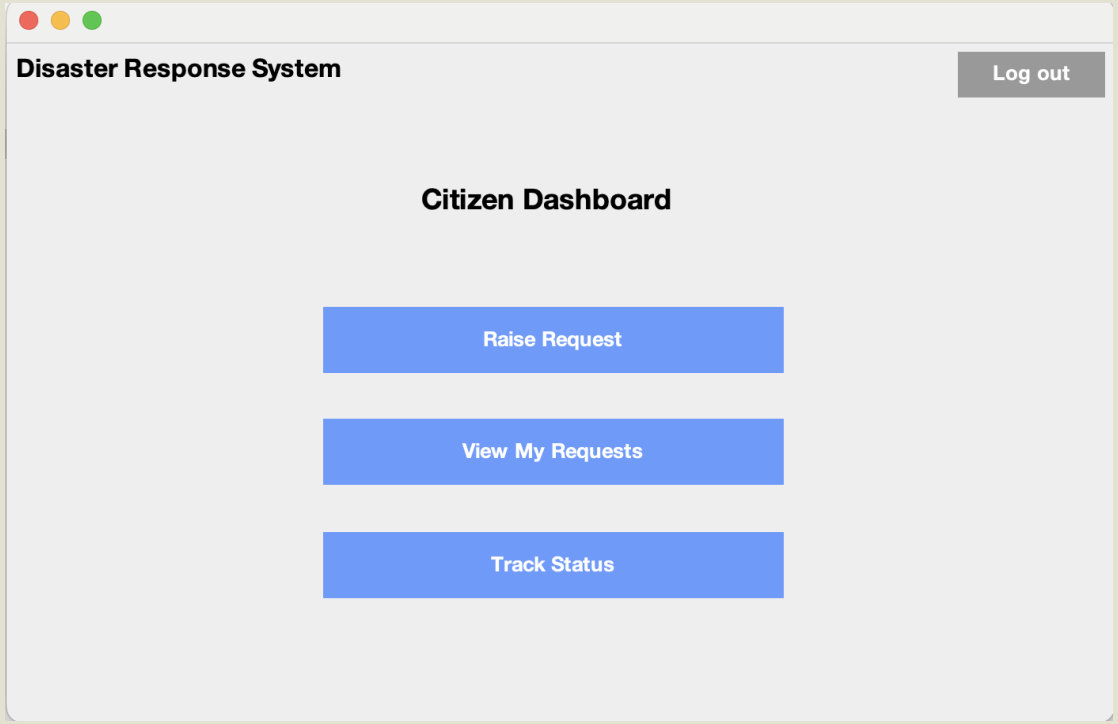


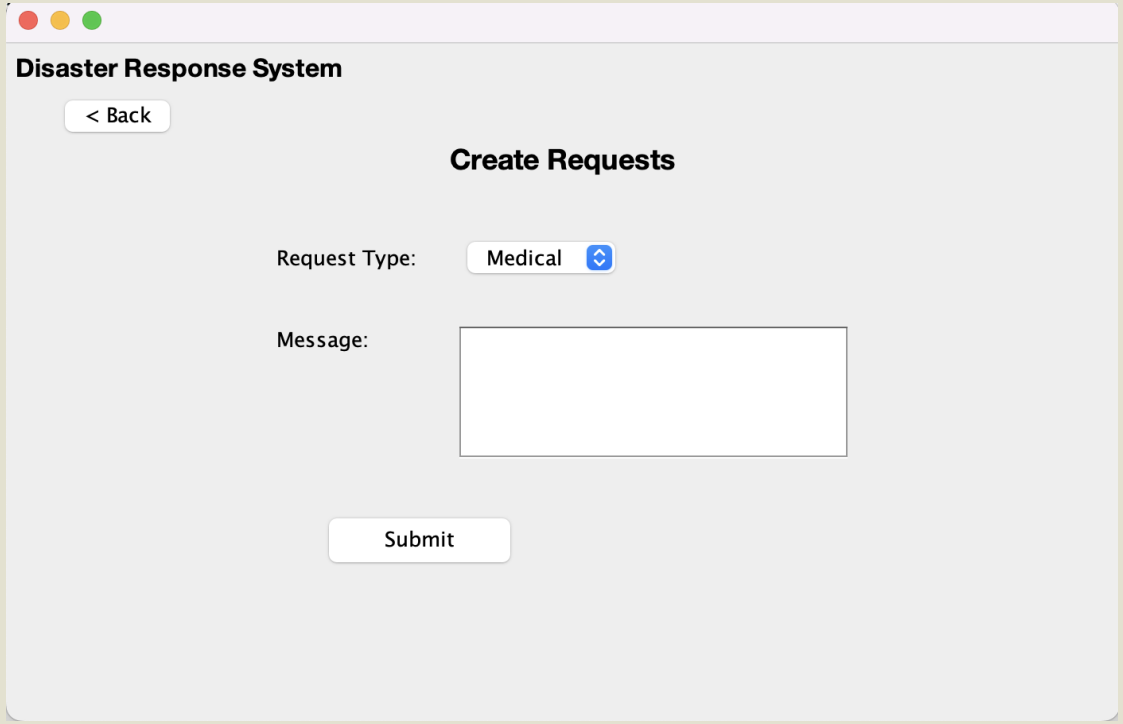


AdminPanel – Manages the system, controls users, and checks overall operations.



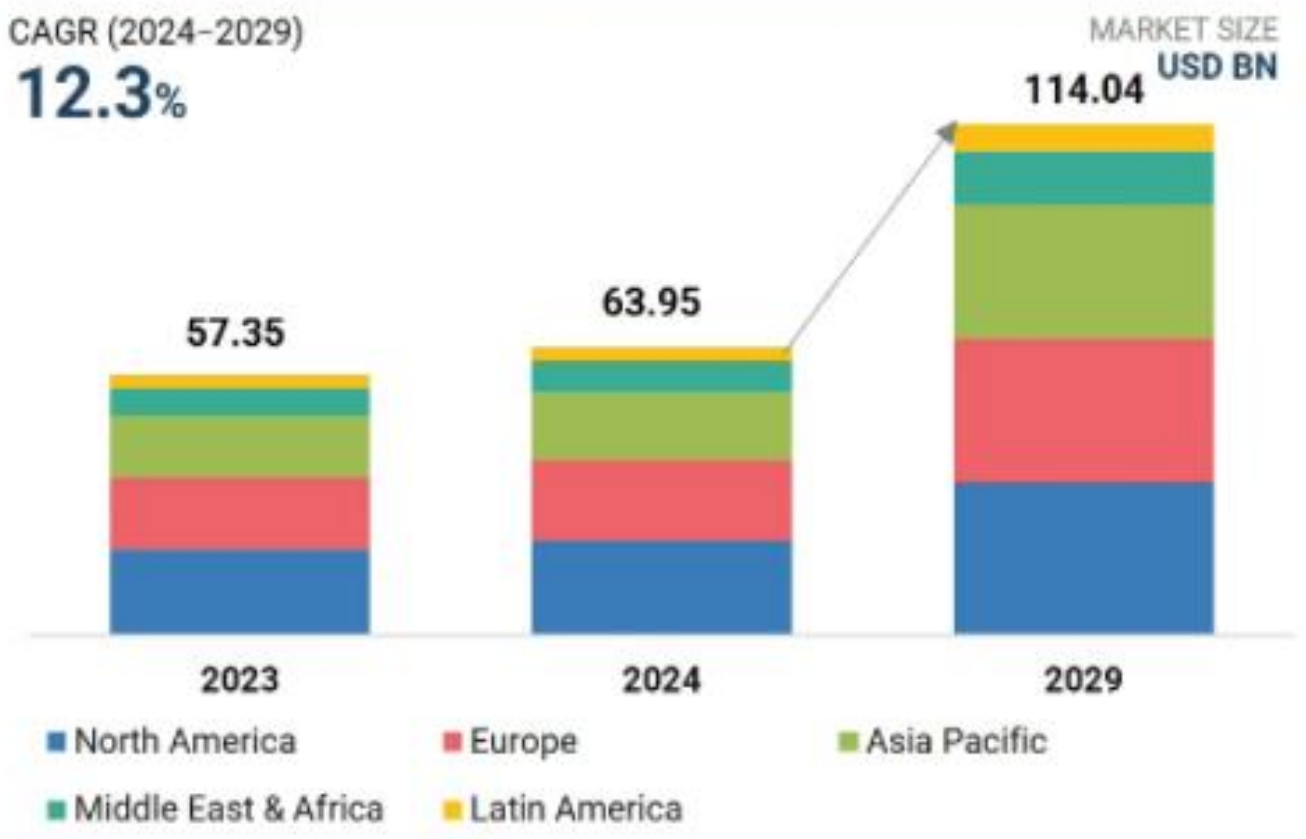
Citizens Raising Requests to different Organizations:





Market Analysis

NATURAL DISASTER MANAGEMENT: MARKET SIZE & SHARE



MARKET SNAPSHOT

Market Size in 2024 (Value)	USD 63.95 BN
Market Forecast in 2029 (Value)	USD 114.04 BN
CAGR	12.3%
Years Considered	2019–2029
Base Year	2023
Forecast Period	2024–2029
Units Considered	Value (USD BN)
Fastest-growing Region	Asia Pacific

Future Scope/Advance Features:

- The system can be expanded to support multiple cities or countries.
- AI can be added to predict disasters and suggest early actions.
- Mobile app support can make it more accessible for users.
- Integration with IoT devices can help in real-time data collection.
- More advanced analytics can be added for better decision-making.
- The system can be connected with government databases for faster response.
- Integration with maps and notifications will improve real-time response.
- It allows different organizations to work together smoothly on the same platform.